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Importance of Soil: Testing & Treatment

(A Primer for Class VI -VIII Students)





About the Document

Soil is a crucial and necessary foundation for plant growth and is a supremely important resource of our country. More than 54% of our workforce either directly depends on agriculture for their livelihood, or through allied activities. Therefore, the health of soil affects more than 54% of our workforce! Soil health is fundamental to the food system in the country. It forms the bedrock of agriculture and serves as the vital medium for the growth of food-producing plants. When soils are healthy, they yield higher productivity of nutritious crops that provide nourishment for both humans and animals. It's important to recognize that the quality and quantity of our food are directly connected to the quality of our soils. Therefore, prioritising soil health is essential for ensuring the availability of high-quality and abundant food for all and reducing dependency on other countries for agricultural produce.

The Honourable Prime Minister Narendra Modi has emphasised on teaching students the importance of soil and its health. He has also called for senior secondary students to be trained and involved in doing soil tests to support farmers. This would be done by utilising school laboratories during school vacation times.

In light of this, the Ministry of Agriculture and Farmers Welfare, the Ministry of Education, and the Central Board of Secondary Education have partnered to create this resource for students and teachers to be apprised of the importance of soil health and testing.

This document provides students of Classes VI - VIII with an overview of the importance of soil health, the diversity of soil found within the country and its uses, an introduction to soil health, soil health cards, and the value of testing soil and treating it for improved quality and fertility. This document acts as a resource that students can pursue on their own to learn more about soil initiatives in the country and participate to become enablers of improving soil quality by supporting farmers through soil testing.





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Chapter 1: Why is Soil and Soil Health Important for Our Country?

- Soil is a natural body on the earth in which plants grow. It has formed as a result of various soil forming factor, processes, and differs in nutrient, mineral contents, and properties.
- Healthy soil produces healthy crops.
- Soil is vital for sustainable crop production and the existence of plants and animals
- Soil has different meanings for different individuals. For a farmer, soil is that surface layer of the Earth, which can be ploughed to raise crops for food, fibre and fodder for family and farm animals. For a civil engineer, soil provides the foundation for the construction of buildings, roads, highways, etc.
- Soil health is important for sustainable crop production, and the existence of flora and fauna. Exploitation, destruction and mismanagement of soil can lead to extinction of various plants and animals, including humans.
- Efforts must be made for the maintenance of soil health, which is important for the sustenance of life on Earth.¹

1.1 The Role of Soil in Agriculture

- Soil is the essential foundation of agriculture and plays a crucial role in sustaining life and facilitating plant growth.
- It serves as a reservoir of nutrients, water, and air, creating an ideal environment for plants to thrive.
- Moreover, soil provides the necessary physical support for plant roots, anchoring them firmly and enabling efficient water absorption.
- Without the fertile binding of soil, agriculture would struggle to exist, emphasising the significance of soil as a fundamental resource in sustaining our food systems and nourishing our growing population.

¹<https://ncert.nic.in/vocational/pdf/kegr105.pdf>





- Healthy soil is pivotal in promoting robust crop growth, ultimately contributing to increased food production and bolstering food security for our nation.
- A fertile and well-nourished soil provides the optimal conditions for plants to thrive, resulting in higher yields and healthier crops. Nutrient-rich soil supplies essential elements that plants need to grow, develop, and produce abundant harvests.
- Through soil testing, we can identify nutrient-deficient soils so farmers can enhance agricultural productivity and ensure a more reliable and diverse food supply by cultivating crops in nutrient-dense soil.
- Robust crop growth driven by healthy soil becomes a cornerstone for achieving food security, meeting the nutritional needs of a nation's population, and fostering agricultural resilience in the face of challenges such as climate change and fluctuating market conditions.

1.2 Environmental Functions of Soil

- Soil serves as a natural water filter, purifying water as it passes through. Water undergoes various processes within the soil profile, enhancing its quality and removing contaminants.
- Filtration occurs as water percolates through the soil, which is composed of different particle size such as sand, silt, and clay, creating diverse pore spaces. These pores act as filters, trapping suspended particles, sediment, and larger organic matter present in the water.
- Consequently, the soil acts as a natural filter, eliminating impurities, sediment, and pollutants from the water. The water that emerges from the soil is generally cleaner, with decreased levels of contaminants.
- This natural filtering process offers valuable ecosystem services benefits by improving groundwater recharge, enhancing surface water quality, and supporting effective water resource management.





- Soil is a vibrant habitat for diverse and countless organisms, significantly contributing to biodiversity. Many microorganisms, insects, worms, and plant roots find their home within the soil.
- This diverse variety of organisms plays vital roles in nutrient cycling, organic matter decomposition, soil structure formation, and disease regulation. Microorganisms break down organic matter, releasing essential nutrients for plant growth.
- Soil plays a crucial role in — mitigating climate change. Through a process called carbon sequestration, soil acts as a carbon sink, removing carbon dioxide from the atmosphere and storing it in organic matter. Plants capture carbon dioxide through photosynthesis and transfer a portion of it to the soil through their roots.
- Once in the soil, carbon can be stored for significant periods, ranging from years to centuries. This sequestration reduces the amount of greenhouse gases in the atmosphere and improves soil fertility and structure.
- Sustainable land management practices such as agroforestry and conservation agriculture can enhance carbon sequestration in the soil. By promoting soil health and increasing carbon storage, soil can mitigate climate change and create a more sustainable future.



1.3 Socio-economic Importance of soil

- Agriculture relies heavily on soil health and contributes significantly to the economy. Healthy soils provide the foundation for productive and sustainable agricultural systems.
- Farmers can optimise crop yields and quality by maintaining soil fertility, ensuring a consistent and abundant food supply.
- Moreover, agriculture contributes to the economy through job creation, income generation, and export opportunities.



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- Sustainable soil management practices are vital for maintaining soil health, preserving agricultural productivity, and sustaining the economic benefits derived from agriculture.
- Soil also provides livelihood opportunities for farmers and plays a crucial role in supporting rural communities. Since agriculture heavily relies on soil, the fertility and productivity of the soil directly impact crop yields and profitability.
- Farmers can generate income from selling crops, dairy products, and other agricultural products by cultivating and managing the soil. Soil-based activities also create employment opportunities in rural areas, contributing to the economic vitality of these communities.





Chapter 2: Types of Soils in India

2.1 Why is studying different types of soil critical for us?

- Soil is a loose surface material covering the uppermost layer of Earth's crust and serving as a natural medium for plant growth. It performs a wide range of socio-economic and ecological functions, of which agriculture is just one.
- It has a porous matrix composed of unconsolidated mineral matter influenced by various factors, including its parent material, climate, organisms, and topography, which shape its characteristics over time.
- Soil can come in a variety of forms and characteristics due to genetic and environmental factors. Differences in these can affect soil colour, texture, depth, and even the capacity to store different nutrients. Therefore, the soil can also serve as a reservoir for nutrients and water for crops, providing anchorage and conducive farmland.
- In combination, the soil is made of mineral material, organic matter, water and air, and in the variation in proportions affects the ability of plants to grow. *Hence, studying the different types of soil is critical for us.*



2.2 What are the different types of Soils in India?

The soils of India can be classified into different groups, such as black, red, laterite, alluvial, saline, desert, alkaline, forest, hilly, and peaty and marshy.

Soil Type	Characteristics	Found in States
Alluvial Soil	Alluvial soil develops on alluvium of river basin and river valleys. The soils are	The soil is found in all States of India along the rivers, in





Soil Type	Characteristics	Found in States
	extensively distributed in the Indo-Gangetic plains and Brahmaputra valley. They also occur in the coastal and deltaic regions. These soils are ideal to grow all the crops dominantly grown in the region (wheat, maize, sugarcane, pulses, oilseed etc.) and cultivation of horticultural crops. They are fertile and most productive soils. The soil greatly differs in colour, texture, drainage, presence or absence of sodium salts, etc.	deltas of the east coast and in the river valleys The Soils Indo-Gangetic plains are the best example of alluvial soil in India This soil is mostly found in alluvial plains of Punjab, Haryana, Uttar Pradesh, some part of Rajasthan, Bihar & Assam
Black Soil 	Black Soil is dark black to grey in colour. The black soils are generally clayey, in texture, deep and impermeable. They swell greatly and become sticky when wet in the rainy season. In the dry season, the moisture evaporates, the soil shrinks and develops wide cracks. The soil is rich in bases, lime and calcium. The pH of black soil is ranges between 7.2 – 8.5. It has neutral to slightly alkaline reactions. The soil is deficient in nitrogen, phosphate and organic matter but rich in potash, calcium and magnesium. Black soil ranges from heavy clay (ill-drained) to loams (well-drained). Black soil is also known as “Regur Soil” or the “Black Cotton Soil” as cotton	Maharashtra, Northern Karnataka, Madhya Pradesh, western Andhra Pradesh, and southern Tamil Nadu.



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Soil Type	Characteristics	Found in States
	crop are dominantly grown in the black soil region.	
Red Soil 	Red soils formed from weathered material of metamorphic rocks. Soils can vary from reddish brown to reddish yellow due to their high iron content. It is porous and friable, neutral to acidic in reaction. Nitrogen, phosphate, lime and humus content are very low in this soil. In general, red soils possess some characteristics of good growing sol.	Tamil Nadu, Karnataka, north-east Andhra Pradesh, eastern parts of Madhya Pradesh, Bihar, West Bengal and Rajasthan
Lateritic Soil 	Laterite soil is formed in areas receiving high rainfall with alternating wet and dry spells. The name “Lateritic” has been derived from the Latin word “later” which means brick. This soil is red to reddish-yellow in colour. Heavy rains cause leaching of bases and silica from the surface of the soil. The soil shows acidic character with pH of 5 – 6 and is poor in nitrogen, phosphorus, potash, magnesium and lime. Such soil is porous and well-drained with poor water-holding capacity.	Eastern Andhra Pradesh, Karnataka, Kerala, Madhya Pradesh, Odisha, Assam and Ratnagiri district of Maharashtra
Desert (Arid) Soil	Desert soils are sandy to gravelly in texture, and is found in low rainfall areas, arid climatic condition. These soils are alkaline in nature with pH value nearly 7.8 to 8.7 and is	Rajasthan, some part of Gujarat and Haryana and Leh-Ladakh region (cold desert).





Soil Type	Characteristics	Found in States
	less productive. It is rich in soluble salts, and poor in nitrogen and organic matter content. The physical conditions of the soil are unfavourable as it has low water-holding capacity due to high sand content.	
Forest and Hill Soil 	These soils are found in forest and hilly regions. These soils are coarser texture in upper slopes and finer in lower/valley sides. The texture of the soil depends on the mountain environment where they are found. Generally, these soils are stony, low in bases and slightly acidic in reaction and infertile for the production of crops. The soils found in the lower valleys are fertile.	UTs of Kashmir, Ladakh, Himachal Pradesh, Sikkim, Uttarakhand
Saline and Alkali soil 	Saline soil shows white incrustation of salts (chlorides and sulphates of sodium, calcium and magnesium) on the surface due to high evaporation during summers. It is also called 'white alkali soil'. Such soil is, generally, infertile and poor in drainage. Such soil is formed as a result of saline irrigation water and over-irrigation for a long time, which raises the water table of the soil. Alkaline soil is rich in carbonates and bicarbonates of sodium and is non-porous. It is also called 'black alkali' or 'Usar' soil.	Uttar Pradesh, Punjab, Haryana, Rajasthan, coastal Odisha and Sundarbans region of West Bengal contain large patches of such soils





Soil Type	Characteristics	Found in States
Peaty and Marshy soil 	This soil is highly acidic in nature and black in colour. Excessive wetness of the soil causes decay and degradation of dead vegetation, forming a layer of partially decomposed organic matter, resulting into peaty and marshy soil.	Tamil Nadu, parts of Bihar and Uttar Pradesh

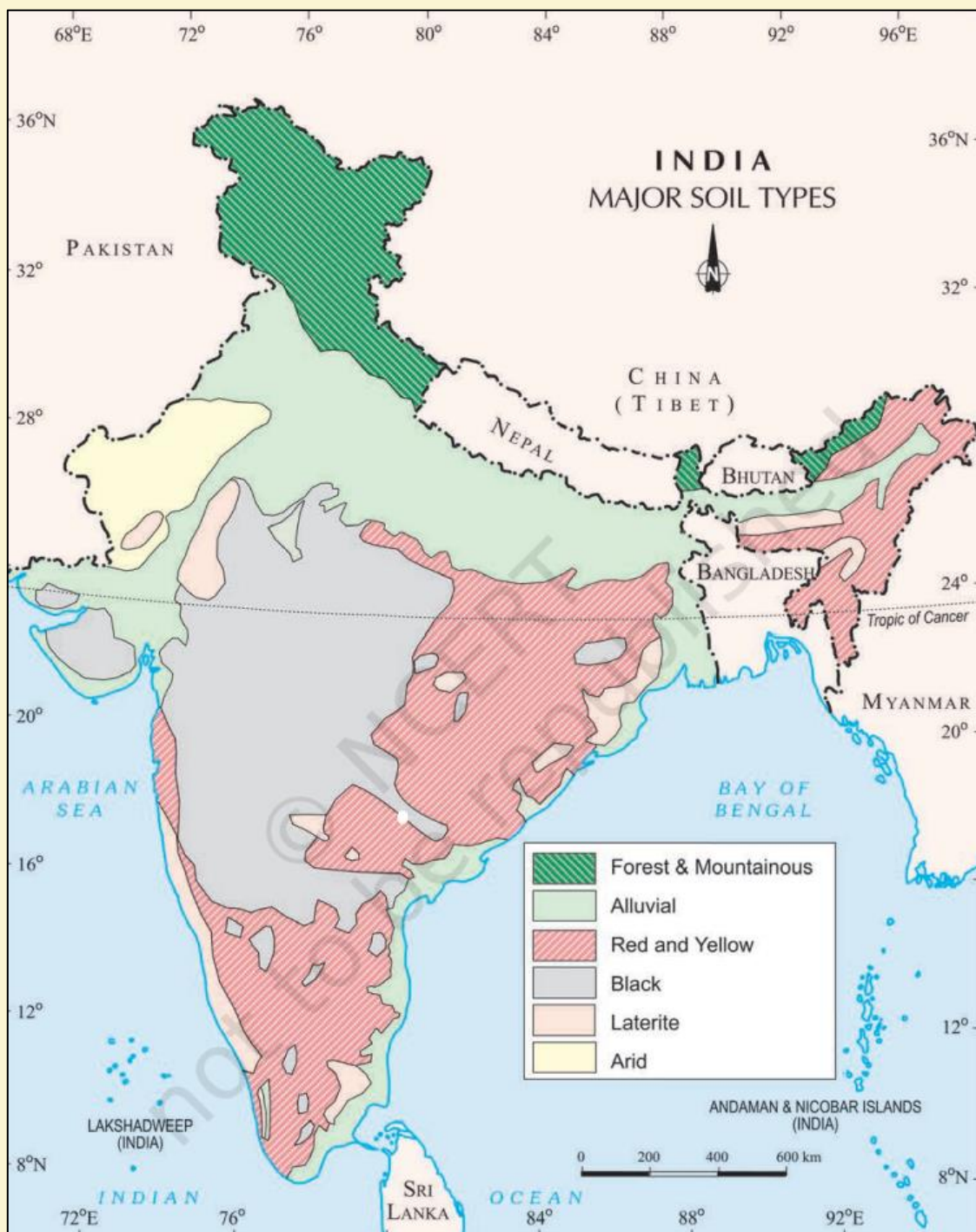
Refer to the following map on the next page to understand soil distribution in India

Fig 1. Soil map of India²

²<https://ncert.nic.in/textbook/pdf/jess101.pdf>



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Chapter 3: What is Soil Science?

3.1 Definition

- Soil science is science dealing with soil as a natural resource on the earth, including Pedology (Soil genesis/formation, classification, mapping) and the physical, chemical, biological and fertility properties of soils to their management and crop production.
- Soil science is a multidisciplinary field combines with biology, chemistry, physics, and environmental science to understand soil properties (both physical and chemical) and processes involved in its formation and subsequent existence.
- The soil science has well defined disciplines of soil fertility, soil chemistry, soil physics, soil microbiology, soil conservation, and pedology.



3.2 Key Concepts in Soil Science

- **Soil composition:** Soils consist of mineral material, organic matter, water, and air, with varying proportions that create a conducive system for plant growth. Thus, it becomes imperative to study soils comprehensively.
 - Parent materials for soil formation primarily originate from rocks, classified into three main types: igneous, sedimentary, and metamorphic.
 - These rocks exhibit significant variations in chemical composition, leading to corresponding differences in soil properties resulting from their weathering processes.
 - Weathering can occur through physical or chemical means. Physical weathering agents include temperature, water, wind, plants, and animals, while chemical weathering involves processes such as hydration, hydrolysis, carbonation, oxidation, and reduction. The interplay of these weathering mechanisms contributes to the diverse characteristics observed in soils.





- **Soil formation or genesis:** The process of transforming rocks into fertile agricultural land is known as soil formation or soil genesis.
 - It is a gradual process that involves the weathering of rocks and subsequent changes that result in the development of arable soil.
 - Five key factors contribute to soil genesis: climate, parent material, topography, plants and animals, and time.
- **Soil properties:** The manner in which soil is formed and, subsequently, the conditions in which it exists dictate its characteristics and properties. These include texture, structure, fertility, pH content, chemical and mineral composition and water-holding capacity.

3.3 Soil properties

Soil can be identified or classified according to various characteristics exhibited by it. The properties of soil are helpful in understanding the nature and kind of the soil. The properties of soil can be categorised as physical, chemical and biological.

a. Physical properties

Soil colour

- Soil surfaces, generally, show black, yellow, red and grey hues. The colour of the soil is due to the parent material, organic matter and minerals present in it.
- The colour of the upper layers of the soil may be different from its other layers.
- The colour of the soil is an indicator of the organic matter present in it, soil fertility status, soil pH, drainage, aeration and organisms living in it.
- The Soil colours are measured with Munsell colour chart and recorded in the form of hue, value and chroma.





- The Munsell soil colour chart gives the dominant spectral colour (hue), degree of lightness or darkness (value), intensity and relative purity of soil colour (Chroma).

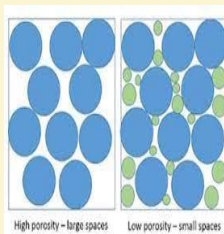
Soil texture

- It refers to the relative proportion of different sizes of particles (sand, silt and clay) that comprise soil.
- Soil, according to the particle size, can be classified as sandy, silty, loamy and clayey.
- Large size particles present in the soil are known as 'sand'. The diameter of sand particles is 0.2–2 mm.
- When the size of soil particles is between 0.2 and 0.002 mm, it is called 'silt'
- Clay is the finest particle, having a diameter of less than 0.002 mm. Loamy and clayey soils have adequate water-holding capacity and are more suitable for the cultivation of crops.



Porosity

- Particles of different sizes are present in soil. When soil particles aggregate, some hollow spaces are formed between them.
- These interparticle spaces in the soil are called 'pores', which help carry air and water to plants and other organisms growing in and over it.
- The quantity and size of pores show the 'porosity' of soil. Soil having a big and large number of pores is called 'porous soil'. Such soil has adequate drainage and aeration capacity.



- Soil having a small but a greater number of pores shows better water-holding capacity. Soil with a small and a smaller number of pores is called 'non-porous' soil. Such soil is stagnant water and is not suitable for cultivation.

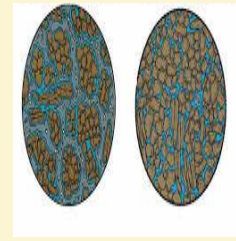
$$\% \text{Pore Space} = 100 - \frac{\text{Bulk density}}{\text{Particle density}} * 100$$

Soil density





- Soil consists of various particles. It has a certain percentage of pore space through which air and water movement takes place.
- The density of soil denotes the weight of the soil per unit volume. It is of two types — particle density and bulk density.
- Particle density: Particle density refers to the actual density of soil solids. It is defined as mass per unit volume of soil solid only. Particle density of soil gets decreased with increase in organic matter content of the soil. Mostly soils have particle density of about 2.65 g/cm^3 .
- Bulk density: It refers to weight per unit volume occupied by soil solids, as well as, pore space of the soil.
- It is expressed as grams per cubic centimetre (g/cm^3). Soils having low bulk densities have better physical conditions.



Soil consistency

- Soil consistency is the potential of soil to change the shape or moulding when moist.
- It also ensures the resistance of soil particles to crushing or pulverising action by implements when dry.
- Soil consistency helps in knowing the tilth. Knowledge about soil consistency is necessary to understand the soil texture in order to perform tillage operations.



Soil structure

- Soil structure denotes varying shapes and arrangements of primary soil particles (sand, silt and clay) into aggregates.
- Soil structure may be columnar, prismatic, blocky, platy laminar, granular or crumb.
- Soil structure can be modified by various soil management practices, such as tillage, manuring, liming, crop rotation, irrigation, etc.





Soil temperature



- Soil temperature is regulated by the Sun and helps in the decomposition of organic material present in it.
- Various factors, such as colour, slope, moisture level and vegetation on the soil affect the soil temperature.

Low, as well as, high soil temperature is harmful for crops. The growth of crops slows down as the temperature falls below 9 °C and ceases when it reaches 50 °C. Microorganisms present in the soil are active when the temperature is 27–32 °C.³

b. Chemical properties

- Chemical properties determine the fertility of soil. These are related to the ability of the soil to supply nutrients to plants growing on or under it.
- Chemical properties of soil depend on the chemical composition of soil particles.
- Chemical properties can be exhibited by soil pH, buffering capacity, electrical conductivity, nutrients released, cation exchange capacity, etc.

Soil pH or acidity of soil

- Soil pH is an indicator of the level of acidity or alkalinity of soil. It shows the potentiality of H⁺ ions.
- It determines the acidic or alkaline reactions of the soil. More hydrogen ion (H⁺) concentration shows the acidic nature of soil, while the concentration of more hydroxyl (OH⁻) ions represents its alkaline nature.
- Neutral condition is produced by an equal concentration of H⁺ and OH⁻ ions. Slightly acidic soils are more suitable for plant growth. Maximum nutrients are available to crops when the pH ranges from 6.5 to 7.
- Soil pH can be measured by pH metre, pH paper method or pH scale. pH scale has a range of 0–14 with 7 as the neutral point, which indicates the equal concentration of H⁺ and OH⁻ ions.

³<https://ncert.nic.in/vocational/pdf/kegr105.pdf>





- As the value decreases, it indicates higher concentration of H^+ ions. Soils with minimum pH are more acidic in nature. Similarly, as the pH goes above 7, the alkaline reaction of the soil increases with the concentration of OH^- ions.

Some of the effects of soil pH on plant growth are:

- In general, soil pH of 6.5 to 7.5 is considered optimum for maximum availability of plant nutrients.
- Low pH (<6.0) results in an increase in aluminium content in the soil. Its excess may be toxic to plants.
- In general, the availability of toxic metals is more in acidic soils. This affects the activity of soil microorganisms.

Buffering capacity of soil

- The capacity of soil that resists sudden change in its pH is called the ‘buffering capacity’ of soil.
- Change in pH may affect nutritional balance in the soil, as well as microbial activities in it. Carbonates, bicarbonates and phosphates act as buffering agents in soil.
- The buffering capacity of the soil also depends on clay and organic matter present in it.

Soil colloids

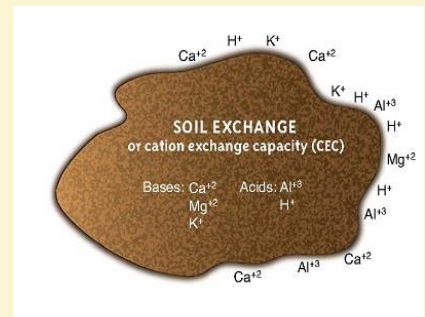
- Soil colloids may be clay or humus. Clay found in soil is known as ‘inorganic colloid’, while humus is called ‘organic colloid’.
- Soil colloids that are charged with negative ions (anions) will attract positively charged ions (cations).
- Cations are held by soil colloids. Cations with lower valency are replaced with ones having higher valency. For example, Na^+ in alkali soils are replaced by Ca^{2+} after the addition of gypsum.
- Cation Exchange Capacity (CEC) of the soil is higher if it has more clay and humus content. CEC is expressed as $cmol (+) kg^{-1}$ of soil





Cation Exchange Capacity (CEC)

- It is a measure of the potential of a soil to hold cation nutrients, such as potassium (K^+), calcium (Ca^{2+}), magnesium (Mg^{2+}), sodium (Na^+), aluminium (Al^{3+}), iron (Fe^{2+}), & hydrogen (H^+).
- In simple words, CEC is the measure of the quantity of cations that can be adsorbed and held by the soil.
- Highly fertile soils, containing high organic matter, have more cation exchange capacity. Soil fertility increases with an increase in cation exchange capacity.



c. Biological properties

- Different types of organisms and microorganisms thrive in soil. Mice, crabs, snails, earthworms, mites, millipedes, centipedes, fungi, bacteria, actinomycetes, protozoa and nematodes are commonly found in soil.
- They feed on plant residues. These organisms make channels and burrows in the soil, thereby, increasing aeration and enhancing the percolation of water due to their activities.
- Their excreta add to the organic matter of the soil. Bacteria predominate neutral soils, while fungi are more in acidic soils. Moist and shady soils favour the growth of algae.





Chapter 4: Importance of Soil Testing and Treatment

4.1 Why should soil be tested?

- To safeguard our soils and maintain soil fertility and productivity, it is necessary to implement holistic soil & crop management practices that foster soil health and ecosystem services.
- These systems should be socially, acceptable, ecologically sound, and economically sustainable, ensuring a balance between human needs and environmental preservation
- Adopting such holistic approaches can protect our soils from degradation while enhancing their productive capacities.
- We must prioritise managing our land and soil resources to secure a sustainable and resilient food future.⁴



4.2 How can soil be tested?

Soil Health can be tested by Soil Testing. Soil testing helps ascertain the status of various nutrients, soil fertility level, pH, etc. It is important to know the fertility status and physical properties of a soil for maximum production and rational soil management.



⁴<https://www.fao.org/soils-2015/news/news-detail/en/c/277682/>





4.3 What are the benefits of Soil testing?

1. Nutrient Management

- Soil testing helps determine the nutrient content and pH status of the soil.
- By knowing the nutrient status of the soil, farmers can apply fertilisers and soil amendments more precisely, matching the specific needs of the crops or plants being grown.
- This improves nutrient uptake, minimises overuse of fertilisers, and reduces the risk of nutrient imbalances or deficiencies.



2. Cost Efficiency



- Soil testing helps optimise fertiliser and nutrient management, leading to cost savings for farmers.
- By applying fertilisers based on the soil test results, farmers can avoid unnecessary or excessive fertiliser applications, reducing input costs.

c. It ensures that fertilisers are applied in the right amounts and at the right time, improving cost-effectiveness and reducing environmental impacts.

3. Environmental Protection

- Soil testing promotes environmental stewardship by minimising nutrient runoff and leaching into water bodies.
- Excessive use of fertilisers can contribute to water pollution, causing eutrophication and harming aquatic ecosystems.
- By applying only the necessary nutrients as indicated by soil testing, the risk of environmental contamination is reduced, promoting sustainable and responsible farming practices.



4. Crop Yield Optimization

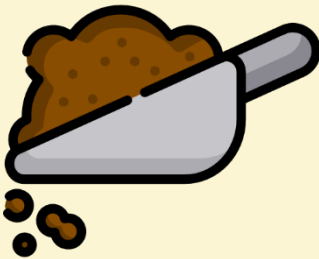




- a. Soil testing provides valuable information about the nutrient availability in the soil, enabling farmers to adjust their management practices to maximise crop yields.
- b. By ensuring that crops receive the required nutrients at the appropriate levels, farmers can optimise plant growth, increase yields, and improve the quality of harvested produce.



5. Soil Health Assessment



- a. Soil testing helps assess the overall health and fertility of the soil. It provides insights about soil organic matter content, soil structure, and other soil properties that impact plant growth.
- b. Regular soil testing allows farmers to monitor changes in soil health over time, identify areas of concern, and take appropriate measures to improve soil quality and long-term productivity.

6. Customised Recommendations

- a. Soil testing results are accompanied by customised recommendations for soil amendments, fertilisers, and pH adjustments based on the specific requirements of the crops or plants being grown.
- b. These recommendations are tailored to address the unique characteristics and deficiencies of the tested soil, providing precise guidance for effective soil management.





7. Sustainable Land Management

a. Soil testing is an essential component of sustainable land management practices.

b. By understanding the nutrient status of the soil, farmers can implement targeted measures to maintain soil fertility, prevent nutrient depletion, and reduce reliance on chemical inputs.

c. This promotes long-term soil health, reduces environmental impacts, and supports sustainable agricultural systems.

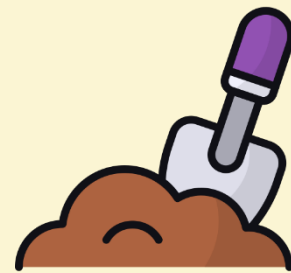
4.4 Steps of Soil Testing

A complete soil test programme essentially consists of three basic steps:

a. Soil sampling

Soil tests and their interpretations are based on the collection of soil samples and their analysis. Therefore, soil samples should be representative of the whole field.

To obtain information about the nutrient status of a soil, it is essential to follow the correct procedure of soil sampling.



- Before sampling information regarding the cropping pattern, various management practices, variations along with the direction of slope, soil colour and texture must be noted down.
- Then, the field from which the sample is to be collected must be divided into 4-5 different sections from centre of the field to other four directions as representative of entire field, and separate samples must be collected from each section.
- By using various sampling tools, such as soil auger, soil tube, spade, etc., the sample must be collected from plough depth, i.e., 0-15 cm for normal agronomic crops and from deeper zones, i.e., 15 – 30 cm for deep-rooted and horticultural crops at different spots,





- Due care should be taken that sample should not take from farm boundaries/ borders, near fertilizer/manure dumps, crop residues etc. below trees, low lying – slopy areas.
- These 4 to 5 soil samples taken from different is then spread on a clean sheet and mix thoroughly. Two opposite quarters are rejected and samples from the other two are mixed. To obtain the desired size of the one composite sample (500 g), the same procedure is repeated.
- The composite sample filled in the plastic or cotton bags. The bags, containing the samples, must be labelled and sent to the nearest soil testing laboratory, along with an information sheet having the following details:
 - Name and address of the farmer.
 - Identification mark or number of the field
 - Date of sampling
 - Local name of the soil, if any
 - Colour of the soil (dry and moist)
 - Type of land (unirrigated, irrigated or waterlogged)
 - Source of irrigation (canal, well or tank)
 - Depth of sampling
 - Topography (level, sloppy or undulated)
 - Crop rotation followed
 - Previous crop
 - Next crop to be taken
 - Manures or soil amendments applied earlier
 - Any other remark

The help of a village-level or extension worker can be taken to collect the soil samples and fill in the information sheet.





b. Soil Testing/Sample Analysis

The collected sample is analysed by using a standardised method in a laboratory for the following parameters:

- pH (indicates whether the soil is acidic, alkaline or neutral in nature)
- Presence of total soluble salts (determined by EC, which indicates whether the soil poses any constraint to seed germination and subsequent crop growth)
- Lime and gypsum, if needed
- Organic carbon, which is a measure of nitrogen content in the soil
- Phosphorus content in the soil
- Potassium content in the soil



c. Soil test interpretation

Based on the soil analysis, data can be interpreted with the help of ratings as given in the following tables.

Table 1: Rating of soil on the basis of pH (1:2.5 soil – water ratio)

S. No.	Type of soil	Soil reaction (pH)
1.	Strongly acidic	<4.5
2	Moderately acidic	4.6-5.5
3.	Slightly acidic	5.6-6.5
4	Neutral	6.6-7.5
5	Slightly alkaline	7.6-8.5
6	Moderately alkaline	8.6-9.5
7	Strongly alkaline	>9.5





Table 2: Rating of soil on the basis of EC(1:2)

S. No.	Category	Category ECe (dSm ⁻¹)
1.	Normal	<1
2.	Critical for germination	1-2
3.	Critical salt levels for growth of sensitive crops	2-4
4.	Injurious to most crops	>4

On the basis of the soil test interpretations, fertiliser recommendations for each crop may be made.

Table 3: Rating of soil on the basis of nutrient availability (1:2)

S. No.	Nutrients	Low	Medium	High
1.	Organic carbon	<0.5%	0.5- 0.75%	>0.75%





2.	Available nitrogen (N)	<280 kg/ha	280-560 kg/ha	>560 kg/ha
3.	Available phosphorus (P)	<10 kg/ha	10-25 kg/ha	>25 kg/ha
4.	Available potassium (K)	< 110 kg/ha	110-280 kg/ha	>280 kg/ha

4.5 How can soil be treated?

- Soil treatment refers to the process of improving or remedying the quality and condition of soil for specific purposes.
- It involves various techniques and methods aimed at addressing soil-related issues such as, nutrient deficiencies, contamination, salinity/sodicity, erosion, or poor fertility etc.
- Soil treatment can be carried out in different contexts, including agriculture, construction, environmental remediation, and soil reclamation.
- Soil treatment and management can be categorised into three broad groups. These pertain to adding organic matter to the soil's and managing the soil's nutrients, and conservation practices that can sustainably protect the soil's health.

i. Organic matter addition

- Using organic materials such as farm yard manure (FYM) compost and crop residues can significantly enhance soil fertility and structure.
- Organic materials are rich in organic matter, which consists of decomposed plant and animal materials.
- When added to the soil, organic matter acts as a source of nutrients, including nitrogen, phosphorus, potassium, and micronutrients.





- These nutrients are released slowly over time as organic matter decomposes, providing a sustained supply to crops growing in the soil.

ii. Nutrient management

- a. Soil testing guides the proper application of fertilisers based on nutrient levels. It informs farmers about soil nutrients status and pH, enabling for fertilization decisions.
- b. Application of fertilisers and fertilization based on test testing results prevents overuse, avoids imbalances, pollution, and economic loss.
- c. If nutrients are sufficient, excess use of fertilisers may be avoided to reduce cost of cultivation and environmental contamination risks.

iii. Conservation practices

- For soil and water management, suitable soil conservation practices should be adopted for preventing soil erosion and land degradation.
- Suitable crop management practices such as crop rotation, mixed cropping, cultivation across the slopes, contour farming etc., are essential for preserving soil health and degradation. These practices work together to minimise soil loss and its soil quality.

iv. Utilising erosion control measures (e.g., terrace/ contour farming)

- helps prevent the loss of topsoil due to water or wind erosion.
- By reducing soil erosion, these practices protect the valuable layer of nutrient-rich topsoil, vital for supporting crop growth.
- Retaining topsoil helps maintain soil fertility and ensures the availability of essential nutrients for crop healthy crop growth.





v. **Water management practices**

- Such as irrigation management and water conservation techniques, also play a vital role in maintaining soil health.
 - Proper irrigation practices help to prevent waterlogging and drought stress, which can improve soil-water-air relationship and nutrient availability.
 - Efficient water management practices also minimises runoff erosion and the potential contamination of water bodies with agricultural chemicals.
-
- ✓ Understanding the importance of soil and its health is very crucial for sustainable agriculture, natural resource management and the well-being of our country.
 - ✓ By promoting soil science education and encouraging soil testing and treatments, we can ensure this precious natural resource for long-term productivity, management and conservation.
 - ✓ Everybody must remember that, Remember, "Healthy Soil Leads to Healthy Crops and, ultimately, a" Healthy Nation.
 - ✓ Thus, as long as we prioritize soil testing to, improve soil fertility, its quality and quality and implementation of conservation practices, we can protect and nurture our precious soil so it can nurture us!





Chapter 5: What is the Soil Health Card?

5.1 Definition and Purpose

The Soil Health Card (SHC) is a tool (in the form of a printed report card) that assesses soil nutrients status and reports its health based on 12 soil parameters. The SHC can also provide information to the farmers with advice about crop-based fertilisers application and possible amendments that can be made to their agricultural soil.

With information about the present soil nutrient status and overall soil health, along with customised advice for improving soil health; farmers are equipped with valuable information to improve their agricultural practices for sustainable crop production.¹

Test Center		Farmer Details	Sample Information
Name - ICAR-IITR Address - Rajendra Nagar, Hyderabad, Telangana, 500 030 Karir00001-20230424-00006		Name - Tuli, D22(R217) Address - Greater Hyderabad Municipal Corporation South Zone, Hyderabad, Bahadurpura mandal, Hyderabad, Telangana, India Mobile - +91*****663	Selected Crop - paddy Plot Size - 1,000 acre Location - 78.39,17.33 SurveyNo - Sampling Date - April 24, 2023 Testing Date - April 24, 2023 at 4:00 PM
pH 7.74 Range 6.5 - 7.5 Recommendation :	Zinc 0.89 mg/kg Range > 0.50 Recommendation :	Scale High, Acid, Saline, Highly Alkaline Low (OC,OM) ● Low, Deficient ● Medium (OC,OM,mbc) ● Medium, Neutral, Sufficient, Alkaline, Slightly Acidic, High (OC,OM,mbc) ● Not selected ●	
EC 0.32 dS/m Range < 1 Recommendation :	Boron 0.69 mg/kg Range > 0.50 Recommendation :		
Organic Carbon 0.78 w% Range 0.50 - 0.75 Recommendation :	MB C Not selected		
Organic Matter 1.34 w% Range 0.50 - 1.20 Recommendation :	Fungus Not selected		
Nitrogen 279 kg/ha Range 280 - 500 Recommendation : 102.5 kg/ha	Bacteria Not selected		
Phosphorus 23.05 kg/ha Range 22.4 - 50.0 Recommendation : 50 kg/ha	F:B Ratio Not selected	Recommendation Soil Conditioner / Amendment FYM / Compost / Manure (Tonne) - 1.6 Macronutrient Fertilizer (in Kg/Bag) DAP - 44.00 kg MCP - 33.50 kg Urea (45) - 1 bags and 36.90 kgs Micronutrient Fertilizer	
Potassium 329 kg/ha Range 150 - 350 Recommendation : 50 kg/ha			
Sulphur 29.00 mg/kg Range 10 - 20 Recommendation :			
Copper 0.74 mg/kg Range > 0.20 Recommendation :			
Iron 4.97 mg/kg Range > 2.00 Recommendation :			

5.2 Components of a Soil Health Card

- Nutrient levels:** The Soil Health Cards provide information about the status of essential nutrients in the soil, such as nitrogen, phosphorus, potassium, micronutrients, and organic matter. This information is reported to help diagnose fertility status improve the nutrient management of agricultural soil.
- Soil analysis parameters:** The soil parameters for which the agricultural soil is tested include information about Soil pH, electrical conductivity, availability of organic carbon content, macro and micronutrient status,
- SHC Recommendations:** Once the soil has been tested and analysed based on the previously mentioned parameters, the Soil Health Cards provide appropriate recommendations and measures to improve soil quality. These include crop-based fertilizer & manure recommendations for improving nutrient use efficiency, suitable sources and doses of fertilizers for enhancing crop productivity and soil health nutrient management, fertiliser distribution and recommended dosages.

